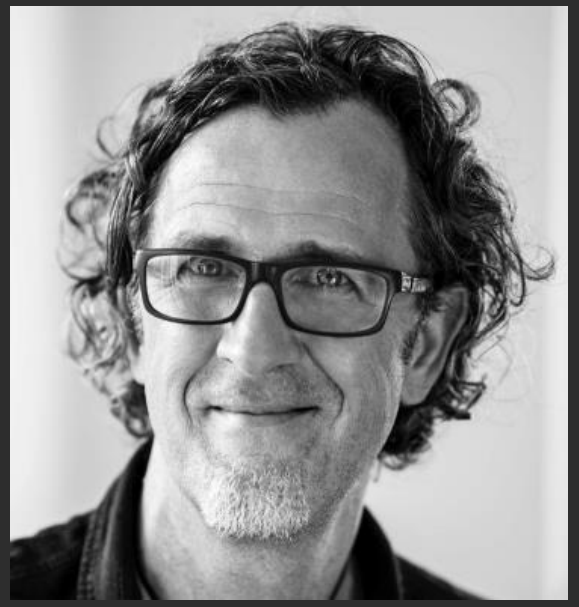
Entra ID Housekeeping -

Identifying and Removing Abandoned Objects to Reduce Risk

About Klaus Bierschenk



Klaus Bierschenk
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Living with my family, two snakes, two cats – and lots of other animals my cats bring home from time to time

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https://nothingbutcloud.net/

Microsoft® MVP Most Valuable Professional

“More than 99% of identity attacks can be prevented with basic security hygiene.”
— Microsoft, Digital Defense Report 2025

https://www.microsoft.com/en-us/security/security-insider/threat-landscape/microsoft-digital-defense-report-2025

So why do we still have stale users, orphaned apps and unused admin roles?

Agenda

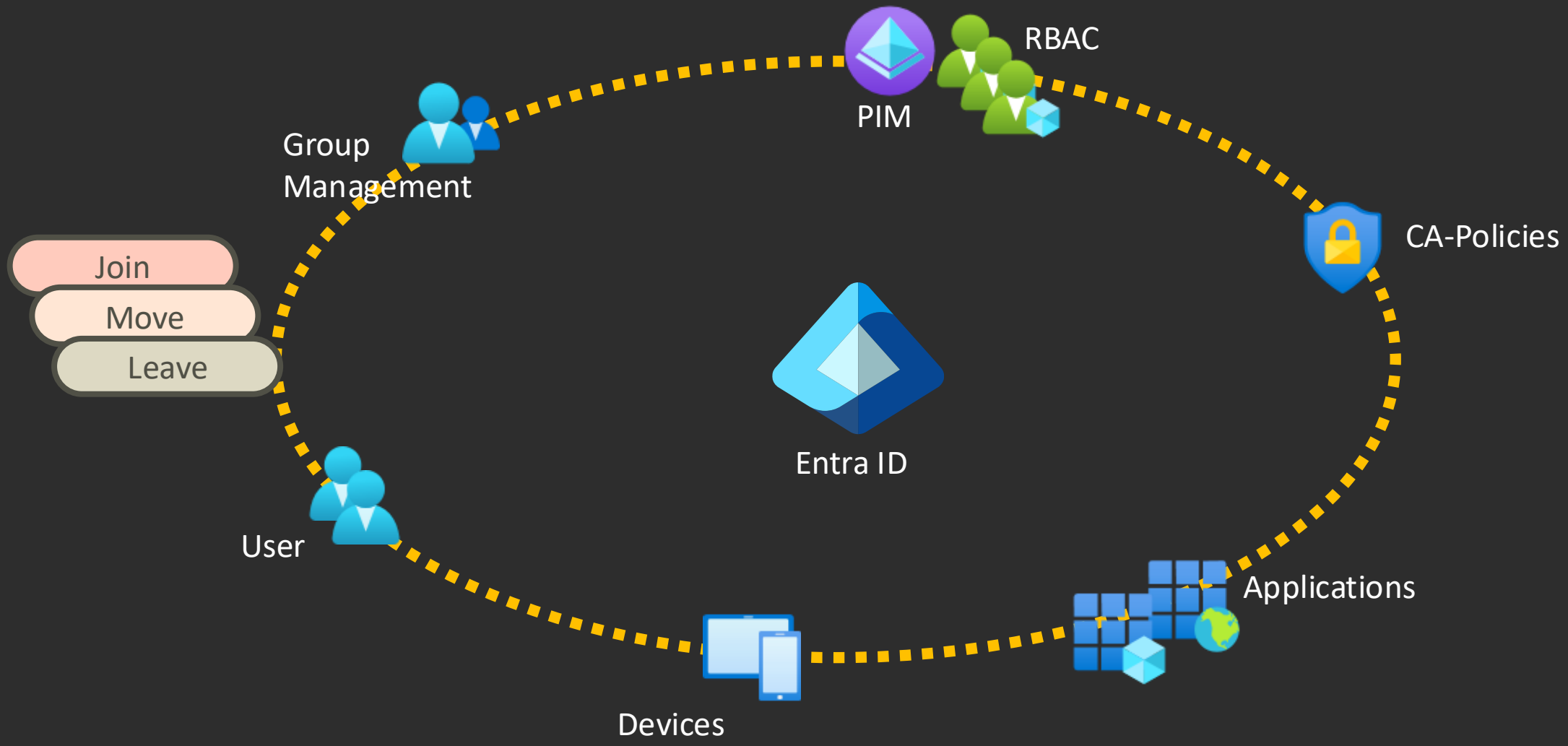
If you don’t clean up your tenant,
someone else might do it for you

- The Causes
  Where and how does tenant clutter arise?
- Tools for Housekeeping
  How do I identify what is stale?
- AI-assisted Housekeeping –
  what's actually useful?

See this symbol?
Referenced Blog Articles exist



How and where does tenant clutter arise?



Does the user still exist?

Characteristics / Risks:

- Quickly created, bypassing standard processes
- Manager or other company attributes empty
- No MFA method
- Is Owner? Manager?

Actions:

- ✓ Entra Admin Center
- ✓ Registered Authentication Methods
- ✓ lastSignInDateTime / PS



- ✓ Lifecycle Workflows
- ✓ Natural Language Query:
  „List all users with no login since 90 days“
- ✗ Access Reviews (rather not)

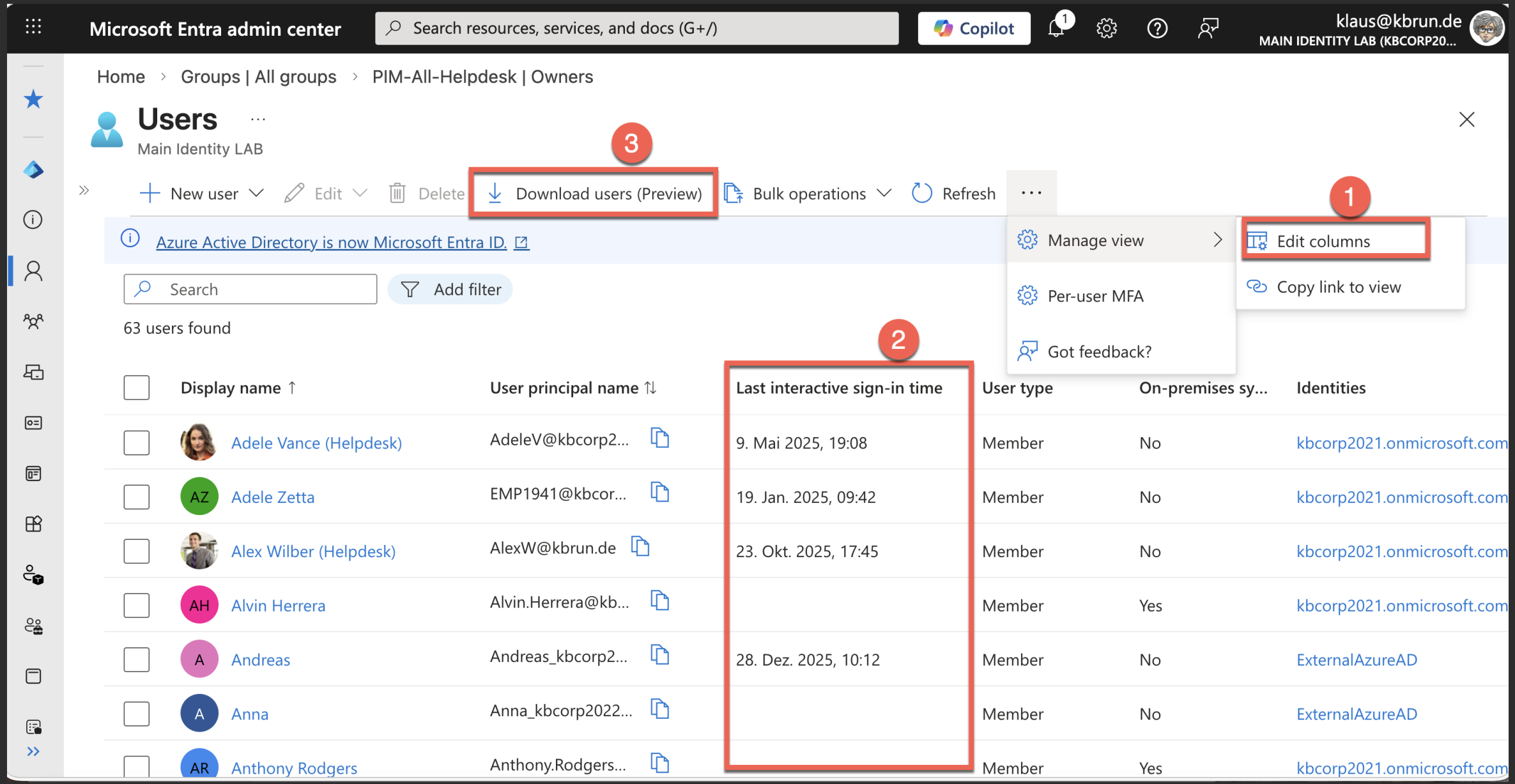
Demo

Entra Admin Center
User Filter

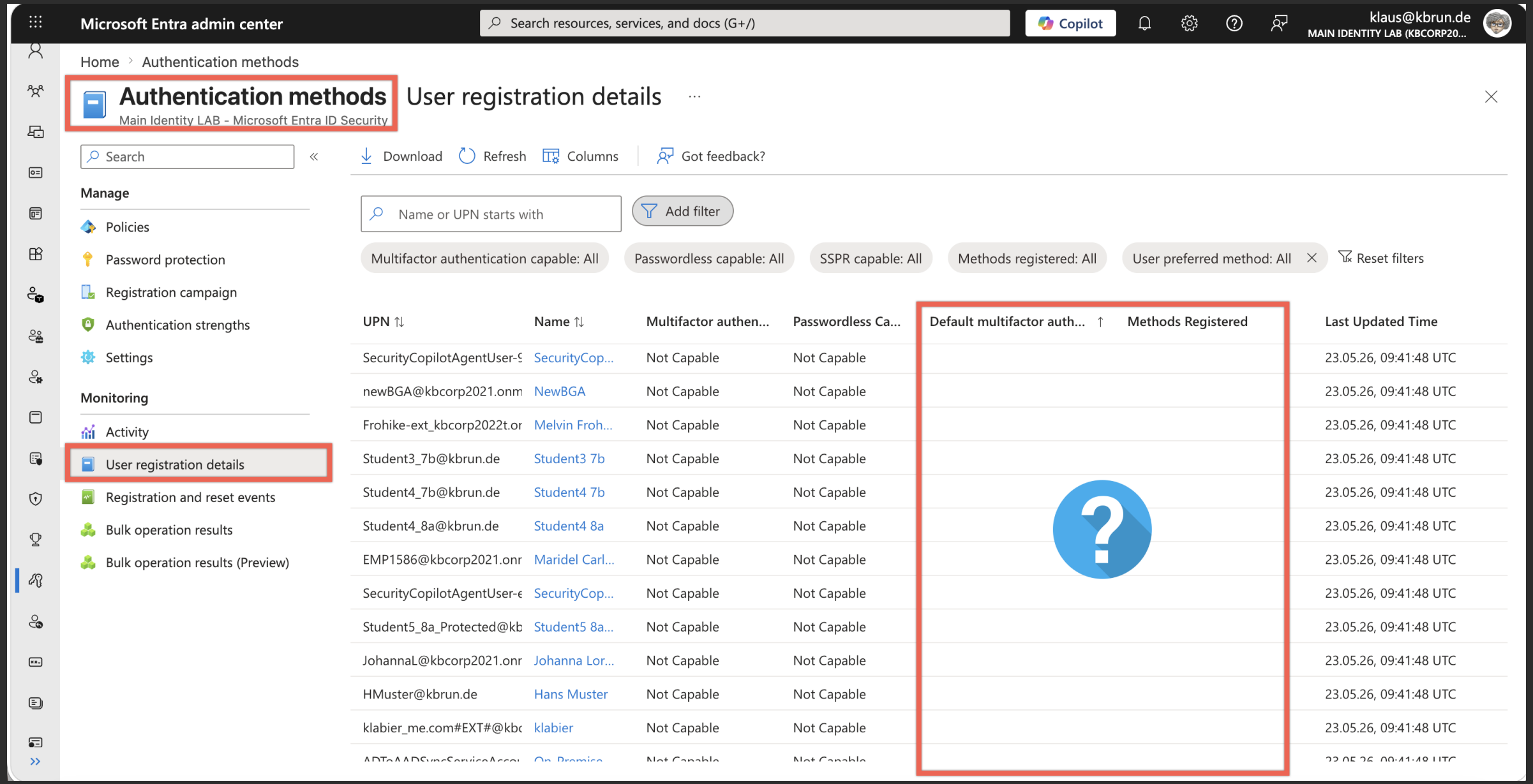
Authentication Methods

PowerShell is your friend





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#NothingButCloud.net

EXPLORER

POWERSHELL

- .vscode
- AAD_Secret_Infos
- AutomationRunbookLab
- BackupCAs_Examples
- Bulk_Operations_CSVs
- CreateSecCopilotSCU
- CSV2SCIM
- Entra - Local Git with EntraExporter
- EntraHouseKeeping
  - 01 User-Beispiele.ps1
  - 02 Groups-Example.ps1
  - 03 Devices-Beispiele.ps1
- SCIMAPIDemo
- CODE_OF_CONDUCT.md
- LICENSE
- README.md
- UnusedObjects.xlsx

OUTLINE

TIMELINE

02 Groups-Example.ps1 | 03 Devices-Beispiele.ps1 | 01 User-Beispiele.ps1

EntraHouseKeeping > 01 User-Beispiele.ps1 > ...

```
27  # The good old SQL days... order by :-)
28  #Get-MgUser -ConsistencyLevel eventual -Count userCount -Filter "startsWith(DisplayName, 'A')" -OrderBy
29
30  # Bestimmte Eigenschaften einschließlich des Datums der letzten Anmeldung
31  #Get-MgUser -All -Property UserprincipalName, Displayname, CreatedDateTime, SignInActivity `
32  #    | Select-Object DisplayName, UserPrincipalName, @{N="Last SignIn";E={$_.SignInActivity.LastSignInD
33
34  # Liste der Benutzer mit Anmeldung vor einem spezifischem Anmeldedatum
35  #Get-MgUser -Filter "signInActivity/lastSignInDateTime le 2025-12-31T00:00:00Z"
36
37  # Liste aller Geräte die länger als 90 Tage nicht benutzt wurden
38  $date = (Get-Date (Get-Date).AddDays(-90) -Format u).Replace(' ','T')
39
40  # ... und das ganze selektiert vor einem Zeitpunkt
41  Get-MgUser -Filter "signInActivity/lastSignInDateTime le $date" `
42      -Property UserprincipalName,Displayname,CreatedDateTime, SignInActivity `
43      | Select-Object DisplayName,
44                      @{N="Last SignIn";E={$_.SignInActivity.LastSignInDateTime}}, `
45                      UserPrincipalName,
46                      CreatedDateTime `
47      | Export-Excel $xlsFile -WorksheetName "Internal User" -ClearSheet -AutoSize
```

PROBLEMS | OUTPUT | DEBUG CONSOLE | TERMINAL | PORTS

```
Writing objects: 100% (10/10), 1.50 KiB | 765.00 KiB/s, done.
Total 10 (delta 6), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (6/6), completed with 4 local objects.
To https://github.com/KlaBier/Powershell.git
   ba0474d..4e806c6  main -> main
PS C:\Users\klaus\OneDrive\GitContent\Powershell>
```

pwsh | PowerShell ...

Powershell | main | 0 0 | Klaus Bierschenk (now) | Ln 8, Col 47 | Spaces: 2 | UTF-8 | CRLF | PowerShell

https://github.com/KlaBier/Powershell/tree/main/EntraHouseKeeping



#NothingButCloud.net


I WANT YOU
To take care of your
Guest Accounts

Is the guest account still needed?

Characteristics / Risks:

- Invitations can happen in many ways
- Without a lifecycle process, immortal
- Source of truth (management) in foreign Tenant
- Admin roles assigned
- CA-Policies applied to guests?
- Sponsor attribute

Actions:

- ✓ Check Guest (Tenant) Settings
- ✓ Who is allowed to invite?
- ✓ Governance – View Inactive Guests
- ✓ Lifecycle Workflows
- ✓ Natural Language Query:
  „list all guest accounts without a sponsor set“
- ✓ Access Reviews!

Demo

Guest Access Governance

Access Reviews

- *For automatic removal*

Lifecycle Workflows



https://entra.microsoft.com/#view/Microsoft_Azure_IdentityGovernance/InactiveGuestsReport.ReactView

Microsoft Entra admin center | Search resources, services, and docs (G+/) | Copilot | klaus@kbrun.de MAIN IDENTITY LAB (KBCORP20...

Home > Main Identity LAB

Insights and reporting | Inactive guest accounts

Microsoft - Entra ID

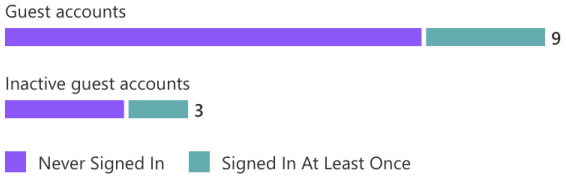
Last updated: 20. Februar 2026 um 02:11 MEZ

Monitor the inactive guest accounts in your organization. Customize the inactivity threshold depending on your organization's needs, narrow down the scope of guests you want to monitor and identify the guest accounts that may be inactive.

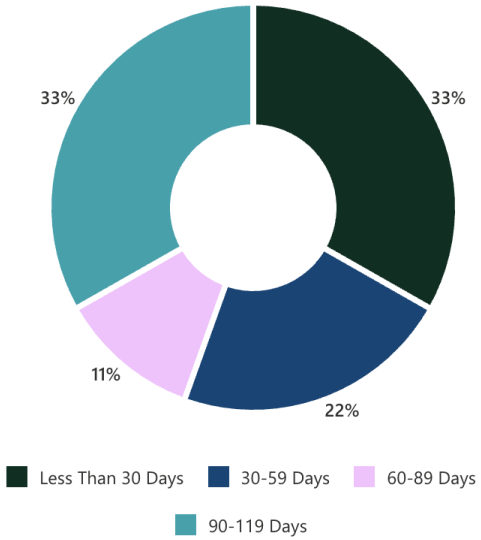
Inactivity threshold

90 days without sign in

Guest account overview

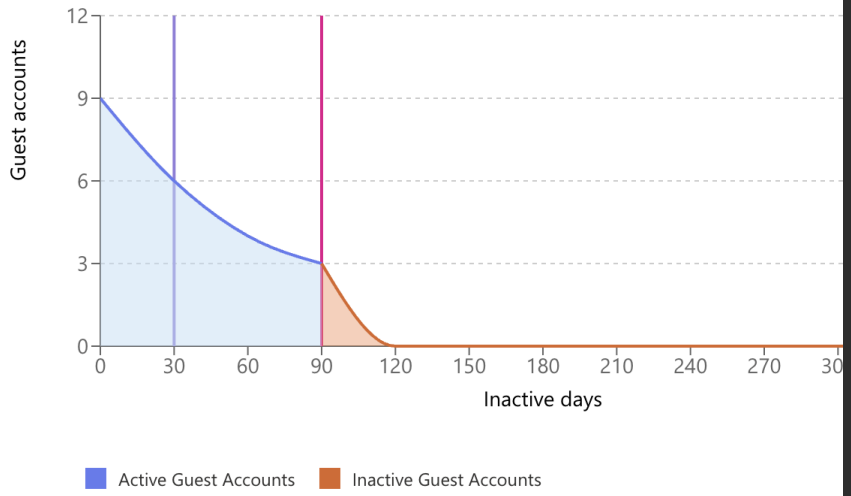


Guests inactivity distribution



Guest inactivity overview

50th percentile	90th percentile	95th percentile	Inactivity thresh...
30 days	90 days	90 days	90 days



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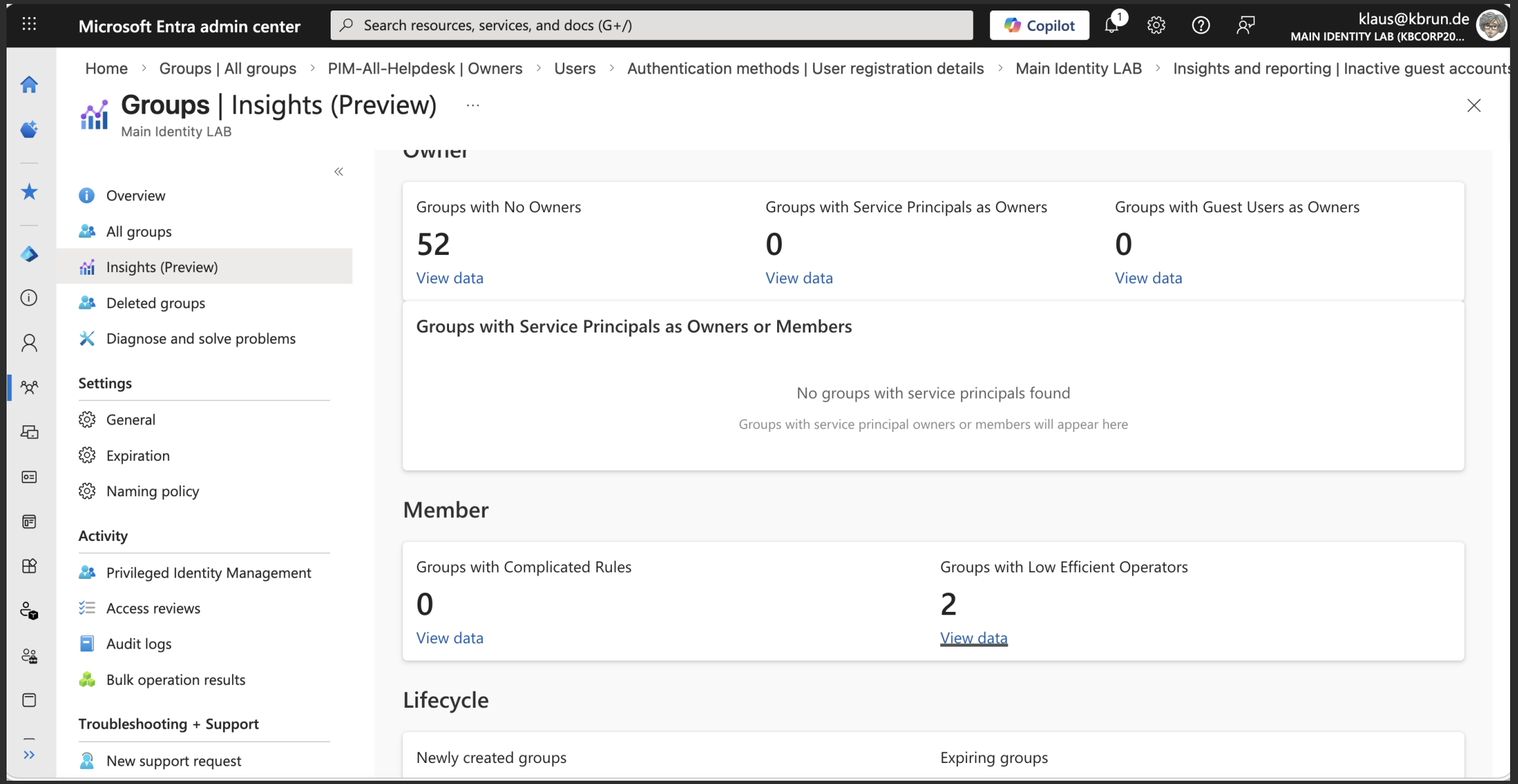
Clean out old groups

Characteristics / Risks:

- Groups without members
- Groups without Owner
- Owner inactive

Actions:

- ✓ Groups Insights (Preview)
- ✓ Powershell / Graph
- ✓ Source of truth?
- ✓ Natural Language Example:
  „list all groups with no sponsor, no member or with any other topics relevant for housekeeping“
- ✓ Managed by vs. Owner



EAC -> Groups -> Groups Insight (Preview)

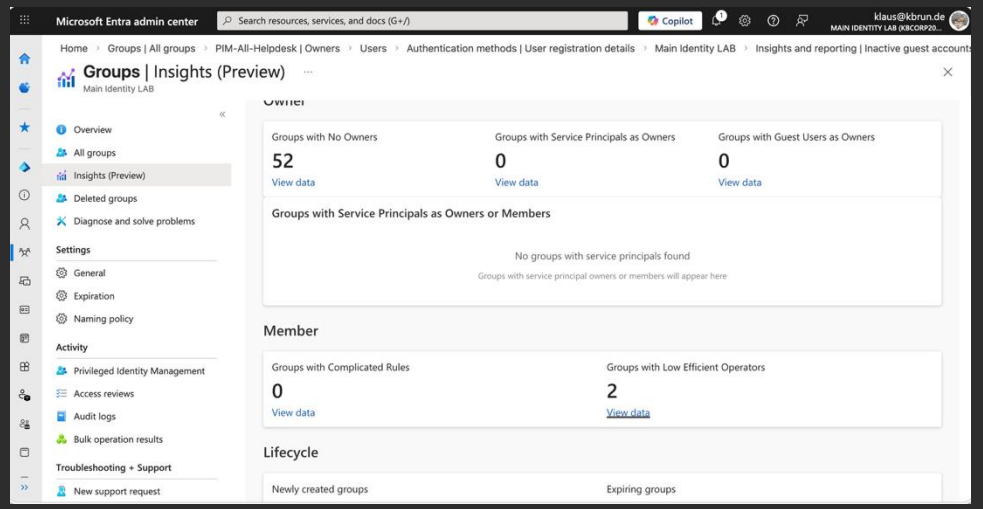
#NothingButCloud.net

Group Insights Feature

- Currently in preview
- Info on 2 levels:

1. Portal UI

   -> Visual overview of group clutter



2. API Endpoint

   -> Data source for custom reports (custom parameters)

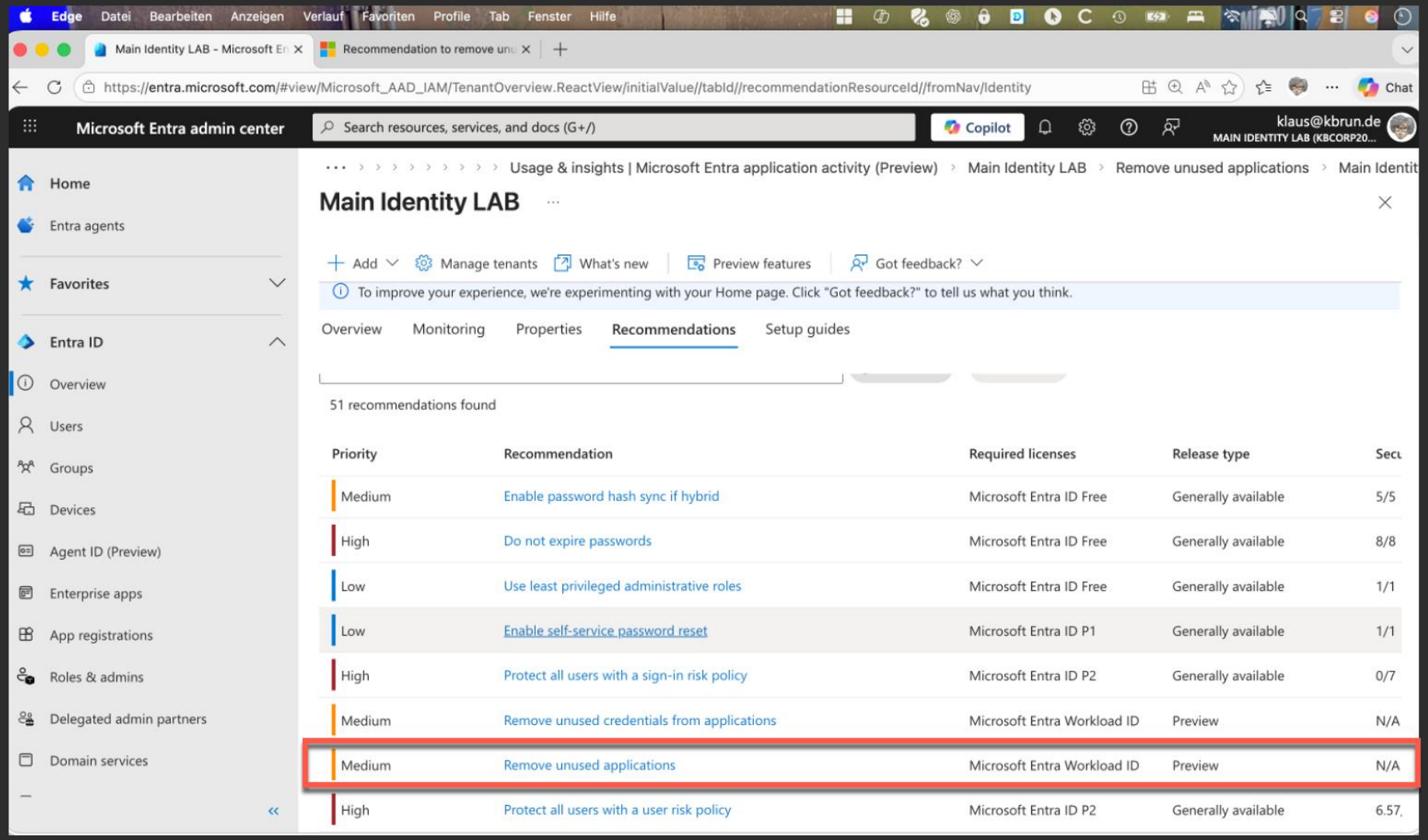
Source: *Jan Bakker – What admins can learn from the new Entra ID Groups Insights blade* (Feb 18, 2026)

Enterprise Apps / App Registrations

Who actually created this?

- No Owner
- No Sign-ins
- AssignmentRequired = Yes
  but 0 Assignments
- Secret Hygiene? (App Registrations)
- Powershell / EAC

Enterprise Apps / App Registrations

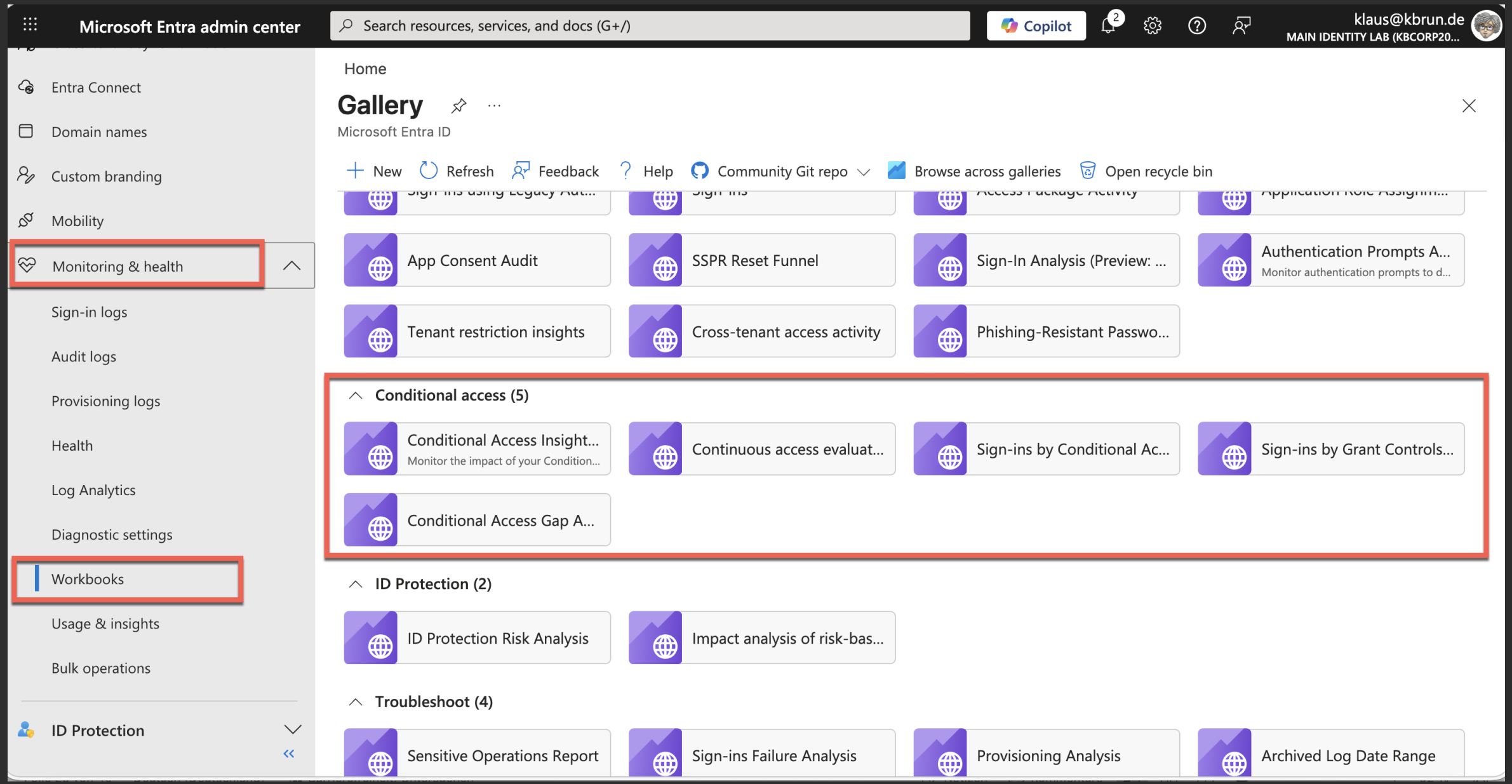


EAC -> Overview -> Recommendations -> Remove unused applications

Housekeeping:
Conditional Access Policies



Admin, looking into his CA
Policies to find unused elements



#NothingButCloud.net

Microsoft Azure | Search resources, services, and docs (G+/) | Copilot | klaus@kbrun.de MAIN IDENTITY LAB (KBCORP20...

Home > Log Analytics workspaces > LA-WS-AAD

LA-WS-AAD | Workbooks | CA Policy Usage

Log Analytics workspace

Help me create my first Azure Workbook | How do enterprises leverage Workbooks? | +1

Search

- Access control (IAM)
- Tags
- Diagnose and solve problems
- Logs
- Resource visualizer
- Settings
- Classic
- Monitoring
  - Insights
  - Alerts
  - Metrics
  - Diagnostic settings
  - Advisor recommendations
  - Workbooks
- Automation
- Help

Add or remove favorites by pressing Cmd+Shift+F

Workbooks | Edit | Help | Auto refresh: Off

CA-Policy Summary

Data coverage

30 days of history available – first entry 30.01.2026

Column Legend
Success = granted
Failure = blocked
NotApplied = conditions not met
ReportOnly = simulation
EnforcedCount = Success + Failure

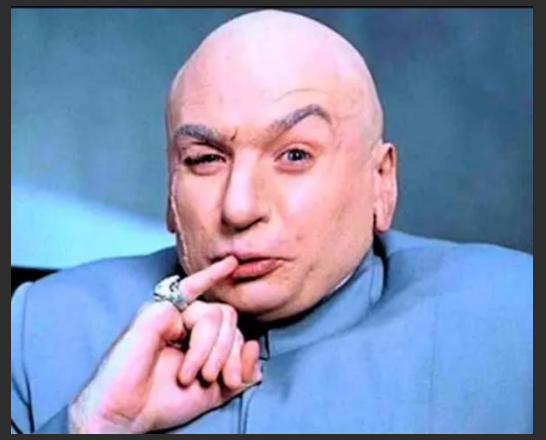
PolicyName	PolicyId	Success	Failure	NotApplied	Report(
CA001-Global-BaseProtection-AllApps-AnyPlatform-Bloc...	cdc3ac53-e78f-45ff-840b-09059df290a6	0	0	0	
CA002-Global-BaseProtection-AllApps-AnyPlatform-Sessi...	acbe4875-9545-4d8c-9ea4-60e1c4d83d56	0	0	0	
CA003-Global-BaseProtection-AllApps-AnyPlatform-MFA	049fab0d-a309-43b9-a3f9-e2f25aa9caf8	455	2	10	
CA003-Global-BaseProtection-AllApps-AnyPlatform-MFA...	c8cfb111-9041-49ed-a46f-c69e1b6c389c	0	0	0	
CA004-Global-EntraTerms-AllApps-AnyPlatform-ToU xxx	0d8db2f1-acb9-4f2f-9fd0-c25230404a8d	0	0	0	
CA007-Global-IdentityProtection-SignInRiskMediumHigh...	b6206cf6-2cc5-4ffe-a5b5-ed447d62682f	0	0	467	
CA008-Global-IdentityProtection-UserRiskHigh-ChangePW	320d5244-2109-4428-b3f6-7eb7e9f59d8d	0	0	467	
CA100-Admins-BaseProtection-AllApps-AnyPlatform-MF...	c4c15e2f-b6d4-48a3-acd6-33099b15c414	362	2	103	
CA101-Klaus-BaseProtection-AllApps-AnyPlatform-SignIn...	33a1da8b-0402-459b-b83c-302d56022e3c	0	0	0	

https://nothingbutcloud.net/2026-02-26-CAPolicyWorkbook/



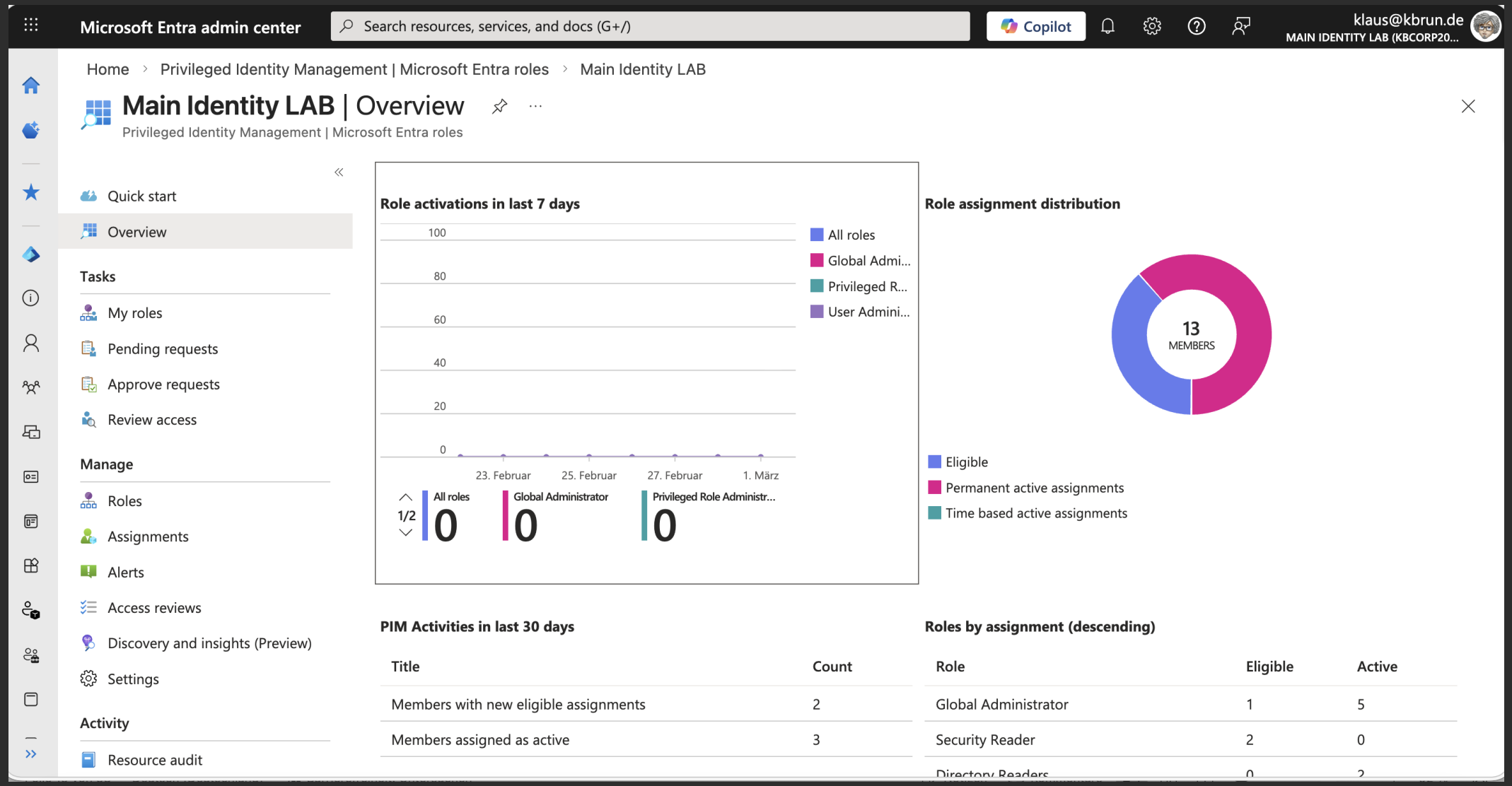
#NothingButCloud.net

Housekeeping:
Privileged Roles - Privileged Identity Management (PIM)



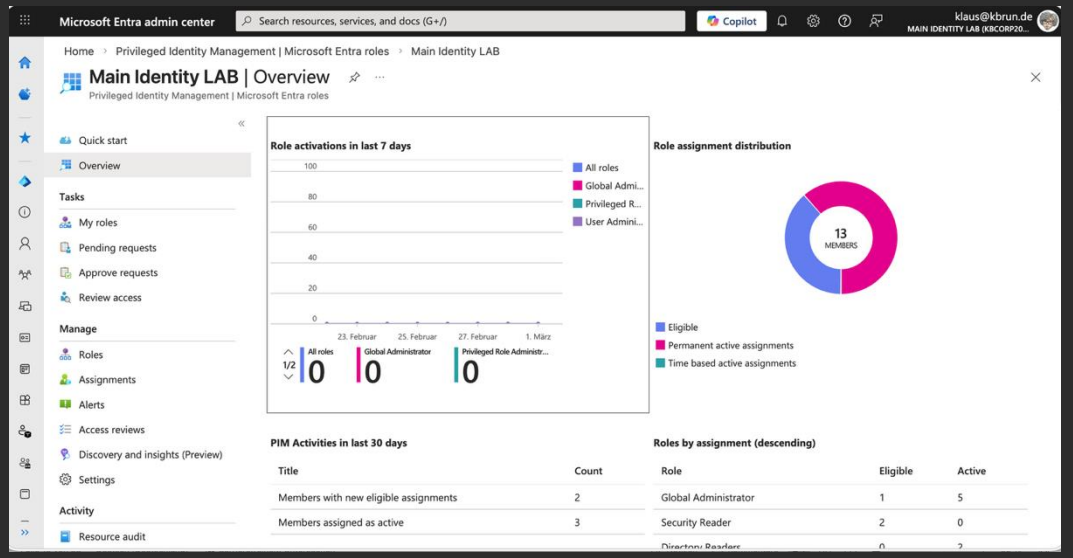
Sure, give everyone Global Admin!
What could possibly go wrong?

Housekeeping - PIM



Housekeeping - PIM

- PIM – Dashboards -> Alerts
- Access Reviews for roles
- Alerts and Access Reviews are related to Privileged Roles
- PIM for Groups? -> Access Reviews (Groups)



Security Copilot …

… and other AI



Much it can do. Know you must.

Security Copilot for Housekeeping?

- Embedded Experience -> embedded in EAC
- Standalone Edition -> securitycopilot.microsoft.com
- Agents: Conditional Access Optimization Agent



- Billed via Security Compute Units (SCUs)
  Computing Units - technical consumption unit, $4 per hour
- Create / remove SCU temporarily via Script

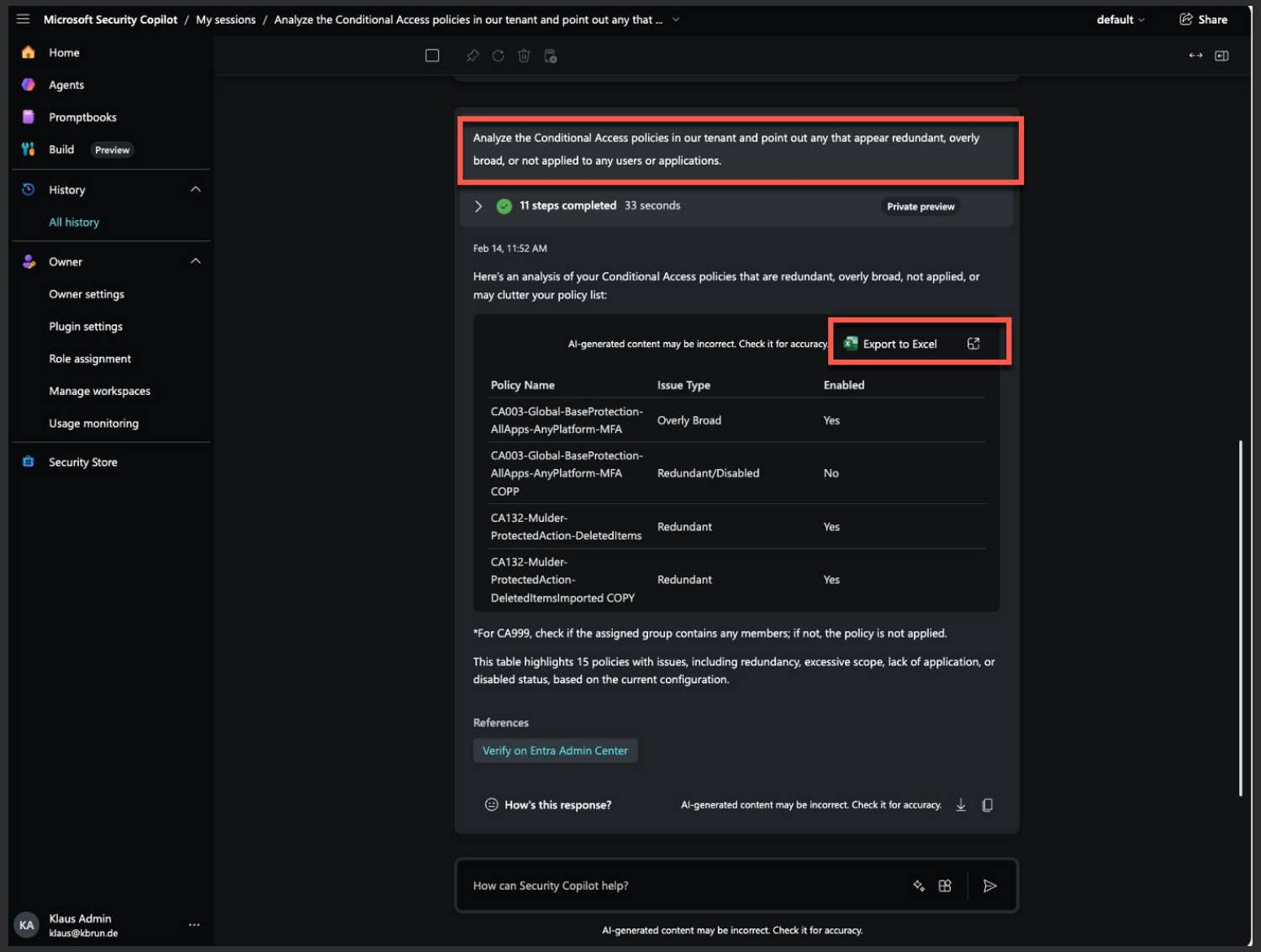


Security Copilot for Housekeeping?

Security Copilot:

Standalone -

Portal / Edition



Security Copilot for Housekeeping?



A1 | Policy Name

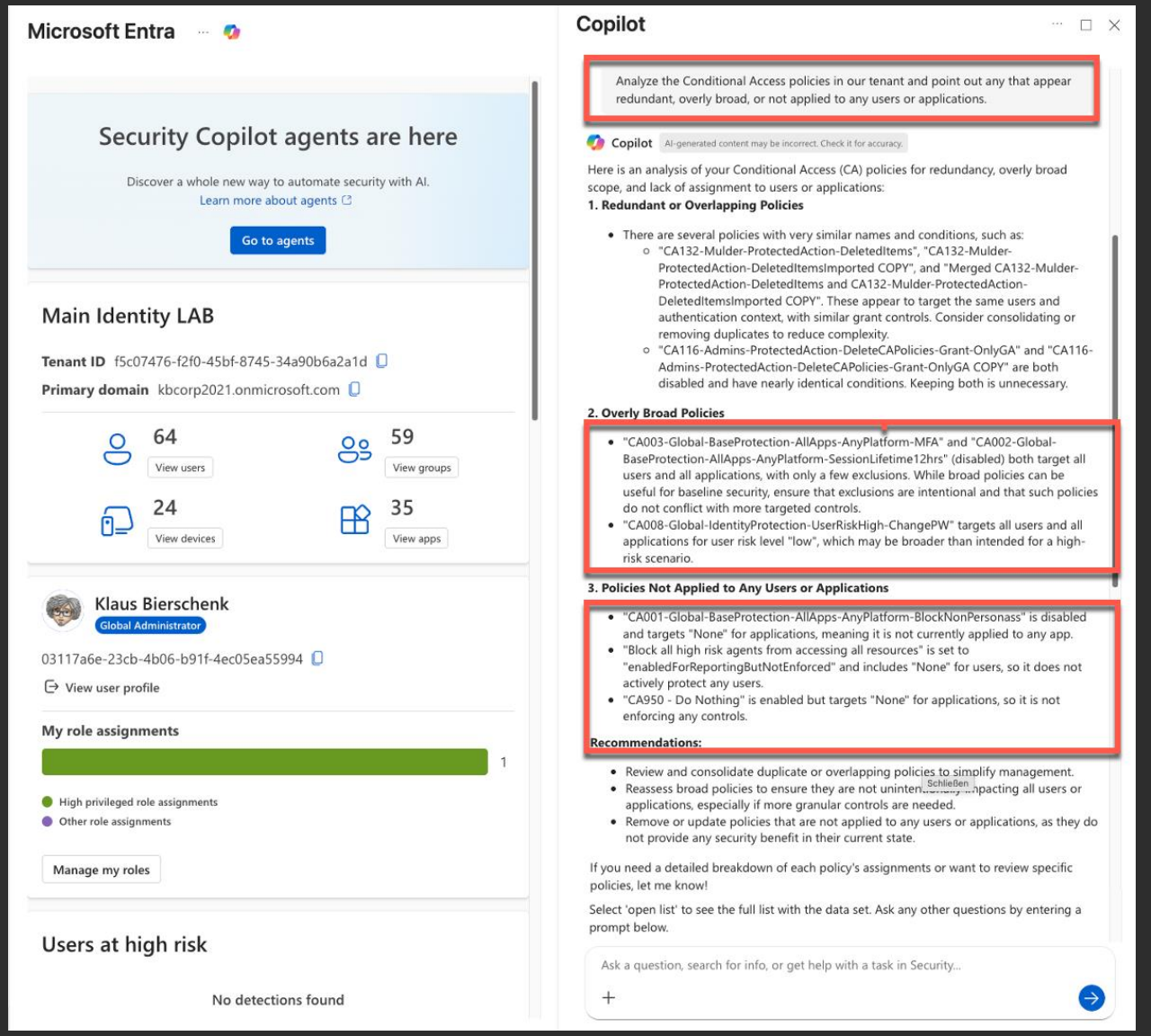
	A	B	C	D	E
1	**Policy Name**	**Issue Type**	**Enabled**	**Target Users/Groups/Apps**	**Details**
2	CA003-Global-BaseProtection-AllApps-AnyPlatform-MFA	Overly Broad	Yes	All users (except 2), all apps (except 1 group)	Applies MFA to nearly all users and apps; may be too broad if more specific policies exist.
3	CA003-Global-BaseProtection-AllApps-AnyPlatform-MFA COPP	Redundant/Disabled	No	All users (except 3), all apps (except 1 group)	Disabled duplicate/variant of above; may clutter policy list.
4	CA132-Mulder-ProtectedAction-DeletedItems	Redundant	Yes	2 specific users, authentication context c1	Nearly identical to "CA132-Mulder-ProtectedAction-DeletedItemsImported COPY"; both target same users/apps.
5	CA132-Mulder-ProtectedAction-DeletedItemsImported COPY	Redundant	Yes	2 specific users, authentication context c1	Duplicate of above; both enabled, same users/apps, same controls.
6	CA950 - Do Nothing	Not Applied	Yes	1 group, no apps	Targets a group but no applications; policy has no effect.
7	CA999-SalesForce Phishing Resistant MFA	Not Applied*	Yes	1 group, Salesforce app	If group is empty, policy is ineffective.
8	CA008-Global-IdentityProtection-UserRiskHigh-ChangePW	Overly Broad	Yes	All users (except 2), all apps (except 1 group), user risk "low"	Triggers on "low" user risk for all users/apps; typically set for "high" risk only.
9	CA004-Global-EntraTerms-AllApps-AnyPlatform-ToU xxx	Disabled/Clutter	No	Not specified	Disabled, may clutter policy list.
10	CA116-Admins-ProtectedAction-DeleteCAPolicies-Grant-OnlyGA	Disabled/Clutter	No	Not specified	Disabled, may clutter policy list.
11	CA101-Klaus-BaseProtection-AllApps-AnyPlatform-SignInFreqNonPersistent	Disabled/Clutter	No	Not specified	Disabled, may clutter policy list.
12	CA002-Global-BaseProtection-AllApps-AnyPlatform-SessionLifetime12hrs	Disabled/Clutter	No	Not specified	Disabled, may clutter policy list.
13	CA888 - Block Langley from Admin Portals	Disabled/Clutter	No	Not specified	Disabled, may clutter policy list.
14	CA132-Mulder-ProtectedAction-DeletedItemsImported COPY (disabled)	Disabled/Clutter	No	Not specified	Disabled, may clutter policy list.
15	CA116-Admins-ProtectedAction-DeleteCAPolicies-Grant-OnlyGA COPY	Disabled/Clutter	No	Not specified	Disabled, may clutter policy list.
16	CA001-Global-BaseProtection-AllApps-AnyPlatform-BlockNonPersonass	Disabled/Clutter	No	Not specified	Disabled, may clutter policy list.
17					
18					
19					
20					
21					

Security Copilot for Housekeeping?

Security Copilot:

Embedded Experience

Entra Admin Center



Security Copilot for Housekeeping?

Eligible Microsoft 365 E5 customers receive
400 Security Compute Units (SCUs) per month per 1,000 E5 licenses (max. 10,000 SCUs per month)

Example 1:
An organization with 400 seats receives 160 SCUs/month

Example 2:
An organization with 4,000 seats receives 1,600 SCUs/month.

- https://aka.ms/securitycopilotme5
- https://aka.ms/SCP-ignite25

MCP Server for Enterprise

- Extend your AI Client to Entra ID for Natural Language Prompting
- No Licensing costs
- Read Only Operations
- Check out Merill Fernando‘s Lokka Project
- Additional reading how to setup
  and additional aspect in my latest blog post



MCP Server for Enterprise

Can you check CA Policies if there a overlapping settings, misconfigs, empty policies etc?

Used MCP Server for Entra Partner Lab integration, loaded tools >

Now let me fetch the full details of each policy to do a thorough analysis:

Used MCP Server for Entra Partner Lab integration >

I now have all the data I need. Here's a thorough analysis:

🔍 Conditional Access Policy Audit

You have **5 policies** total. Here's what I found:

✅ Healthy Policies

CA02-All:MFA-AdminAccounts and **CA03-All:MFA-UserAccounts** are well-structured and complementary. CA02 targets 12 admin roles, CA03 targets all users — together they provide MFA coverage. Both correctly exclude the same break-glass group (`1394bdc7...`).

CA01-All: SessionLifetime12h-AllUser enforces session controls (sign-in frequency: 16h, persistent browser: never) for all users. Note: The name says "12h" but the configured value is **16 hours** — a naming inconsistency worth fixing.

⚠️ Issues Found

1. 🔴 Overlap — Admins get MFA twice

CA02 targets admin roles with MFA. **CA03** targets **All users** with MFA. Since admins are also users, every admin account is covered by *both* policies simultaneously. This isn't necessarily harmful (MFA is still just MFA), but it's redundant and adds evaluation overhead. You should either:

- Exclude admin roles from CA03, or
- Consolidate into a single MFA policy with "All users" scope.

This looks like it may have been created for a specific workflow (step-up auth via authentication context). The naming is confusing and it has **never been modified** since creation (April 2025). Recommend reviewing whether this is intentional and renaming it to reflect its actual purpose (e.g., `CA04-GlobalAdmins:MFA-AuthContext-c1`).

4. 🟡 Session Policy Has No Grant Control

CA01 has `grantControls: null` — it only enforces session lifetime but doesn't require MFA or any other access requirement as a precondition. This is architecturally fine (session policies often stand alone), but means a user could have a long-lived session established without step-up. Evaluate whether adding a compliant device or MFA grant would be appropriate.

5. 🟡 Naming Inconsistency in CA01

Policy name says **"SessionLifetime12h"** but `signInFrequency.value` is **16 hours**. Low severity but misleading for operators.

📋 Summary Table

Policy	State	Issue
CA01 - SessionLifetime12h-AllUser	✅ Enabled	Name says 12h, config is 16h
CA02 - MFA-AdminAccounts	✅ Enabled	Overlaps with CA03 (admins are also "All users")
CA03 - MFA-UserAccounts	✅ Enabled	Overlaps with CA02
AllowDeletedItems	⚠️ Enabled	Confusing name, auth context scope, needs review
test	🔴 Report-only	Empty policy — targets no users, no apps. Delete it.

Recommended Actions

1. **Delete or reconfigure** the `test` policy immediately.
2. **Rename** `AllowDeletedItems` to something meaningful and document its intent.
3. **Fix the name** of CA01 to reflect the actual 16h sign-in frequency.
4. **Consider excluding admin roles** from CA03 to eliminate the MFA overlap, keeping the policies clean and non-redundant.
5. **Review CA01** to decide if a grant control (e.g., compliant device) should accompany session enforcement.

Takeaways

- ✓ Consistently eliminate orphaned identities
  - Users, groups, applications
  - Powershell and automation (Blog Article)
  - Documentation



- ✓ Define a Governance Strategie
  - Access Review
  - Lifecycle Workflows

- ✓ Housekeeping is not a project – it is a process

Microsoft Learn

Embrace cloud-first posture: Convert Group Source of Authority to the cloud resources

Use Microsoft Entra groups to manage role assignments

Manage guest access with access reviews

Create an access review of Azure resource and Microsoft Entra roles in PIM

Best Practices & Community

Jan Bakker – What admins can learn from the new Entra ID Groups Insights blade (Feb 18, 2026)

Merill Fernando - Locca



NothingButCloud Blog

Natural Language Queries Against Entra ID — No Extra License Required ← **Blog Post (Claude and Entra)**

KlaBier Github: CreateSecCopilotSCU PowerShell

KlaBier Github: Powershell Housekeeping examples ← **Examples from today session**



The dilemma with unused objects in Azure AD

Lifecycle Workflows and Custom Extensions - step-by-step-guide

How to Build a Log Analytics Workbook for Unused CA Policies